

ARRAKIS

HARVESTER DOWN

Fly an ornithopter. Rescue the crew.
Escape a pursuing sandworm.

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Roll: 2107096 | Year: 4th | Semester: 1st | Group: B2

CSE - 4102

Computer Graphics and Image Processing Laboratory

Department of Computer Science and Engineering
Khulna University of Engineering and Technology

C++17 + OpenGL 3.3

Procedural models, mathematical motion, game-like controls



Outline

A world, its equations, the player's actions, and the frame that explains them.

01	Build the world	Slides 3-5
	Objects and screenshots; procedural curved surfaces	
02	Bring it to life	Slide 6
	Flight, machinery, workers, pursuit and attack	
03	Give the player control	Slide 7
	Keyboard + mouse, three cameras, winch and delivery	
04	Explain the result	Slides 8-9
	Rendering, feedback, audio and verification	

THE RESCUE LOOP

36 workers

Leave the moving harvester

8-seat cabin

Hover and winch in small loads

Cyan base

Deliver: only then does score increase

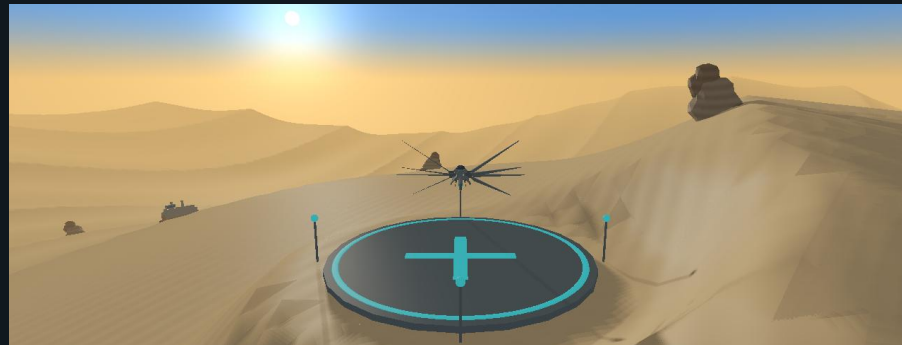
Threat + outcome

Breach, extraction, debrief and retry

Objects I / Desert and machines

Every visible model is generated in code, assembled from indexed meshes, and shaded in real time.

01 / DUNE TERRAIN



900 x 900 patch; 260 x 260 cells.
Height field, smooth normals, flat base terrace.

04 / THREE SPICE TANKS



Cradle/skids, cylinder vessel, domed ends,
bands, hatch/valve and emissive indicator.

02 / ORNITHOPTER



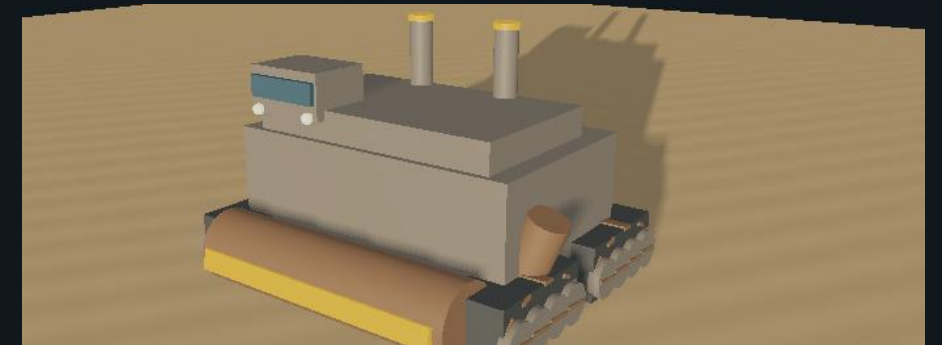
Eight wings; hull/canopy, intakes, gimbals,
turbines/nozzles, tail, sensor and fixed skids.

05 / ROCK FORMATIONS



Eight sites x four pieces: irregular facets,
different scales, strata and procedural grain.

03 / SPICE HARVESTER



Four crawlers, wheels/pads, cutter drum,
exit/ramp, separators, bridge, stacks and lamps.

HOW THEY ARE ASSEMBLED

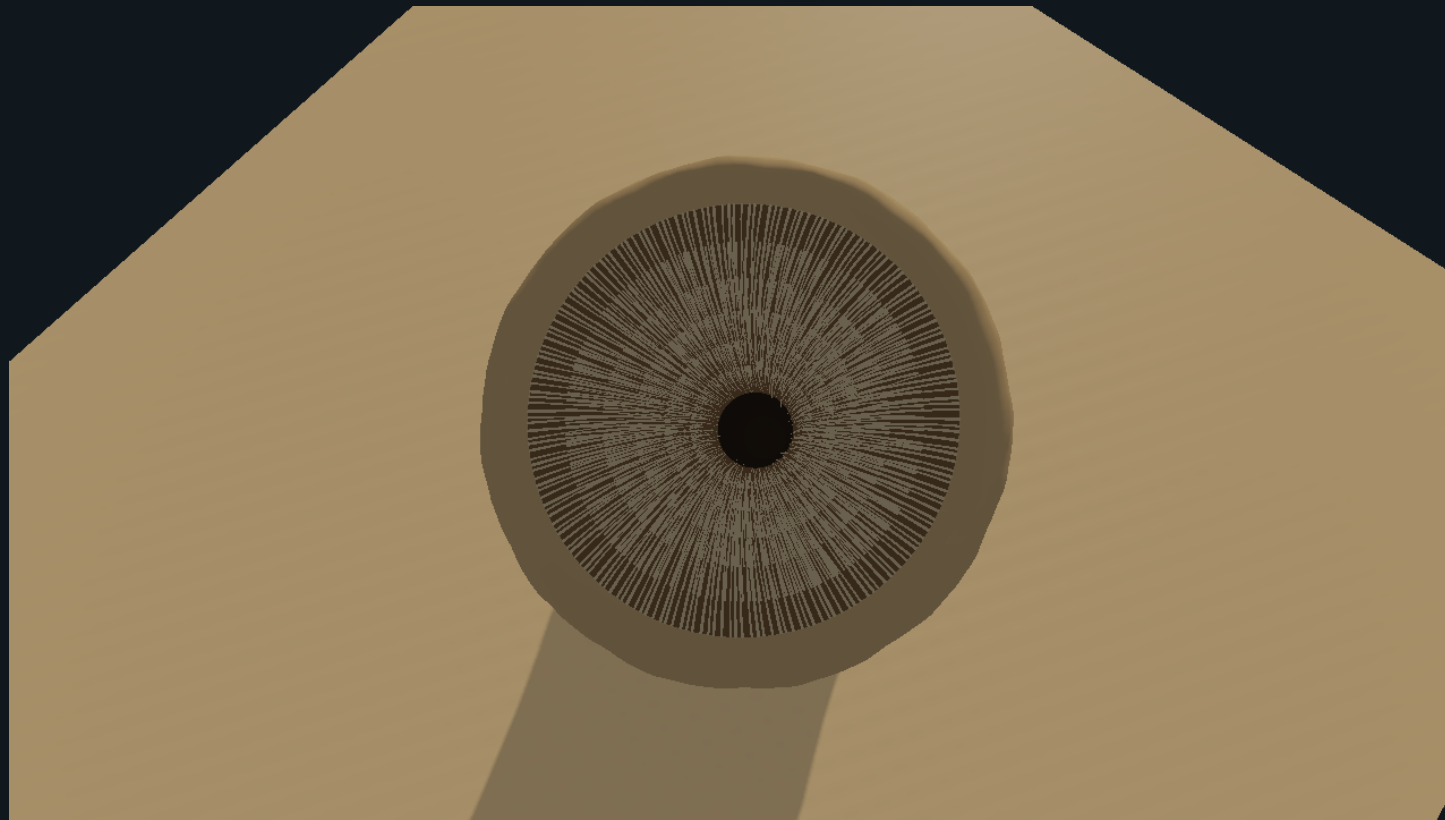


Cube + cylinder + ellipsoid + wing mesh.
Parent/local transforms reuse GPU geometry.

Inspection close-ups reuse actual project draw routines. Terrain is a scene capture. No imported models or image textures.

Objects II / Rescue and sandworm

The mission uses visible people, colour-coded markers, a winch line, and a genuinely recessed worm mouth.



**A modeled cavity, not a black disc.
Body + lip + funnel + deep throat + teeth**

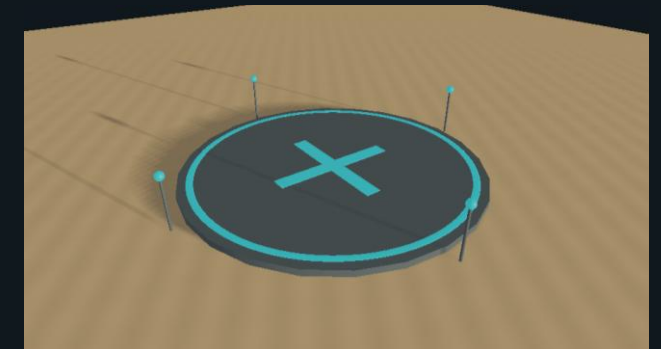
5 meshes | 1,000 curved teeth | 32 body ridge cycles

WORKER



Torso, helmet/visor, limbs, locator

SAFE BASE



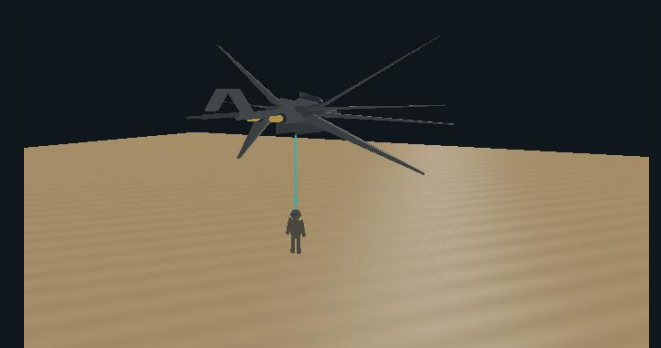
Pad, cyan ring/cross, four lights

AMBER BEACON



Slope-aligned ring, pole and lamp

WINCH + LIFT



Cable + progress-based lift

AMBER: awaiting rescue | CYAN: base / winch | WORM WAKE: four sand mounds + disturbed dust

Curved geometry / The equations behind the objects

Parameters control shape; triangles sample the equations; normals control lighting.

DUNES / a multi-frequency height field

$$H_0(x, z) = 28\sin(.009x + .004z) \\ + 19\cos(.004x - .012z) + 5[1 - |\sin(.028x + .016z)|]^2 \\ + 1.8\sin(.065x + .038z) - 4$$

Broad dunes + sharper ridges + small undulations.
Base flattening: smooth blend from radius 15 to 25.

WORM BODY / bend a radial surface

$$P(t, a) = C(t) + \rho(t, a)(\cos a, -T_z \sin a, T_y \sin a)$$

C(t): buried cubic Bezier + curved 64-unit neck.
t: along body; a: ring angle; T: centreline tangent.
rho: radius + 32 ridges/grain + mouth taper.
Profile revolution creates lip, funnel and throat.

ELLIPSOIDS / CYLINDERS / RINGS

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} + \frac{z^2}{C^2} = 1$$

A/B/C: half-axis lengths; cylinder: unit height, radius 0.5.

$$x^2 + z^2 = .25, \quad -.5 \leq y \leq .5$$

$$P(a, b) = ((1 + .025\cos b)\cos a, .025\sin b, \\ (1 + .025\cos b)\sin a)$$

Torus a/b: major-circle / tube angles; radii 1 and 0.025.

TEETH / a curved, tapered centreline

$$P(t) = R(r - \ell t) + S(ct^2) \\ + (0, d + .65\sin(\pi t) - Dt^2, 0) \\ q(t) = w(1 - t)^{.85}, \quad 0 \leq t \leq 1$$

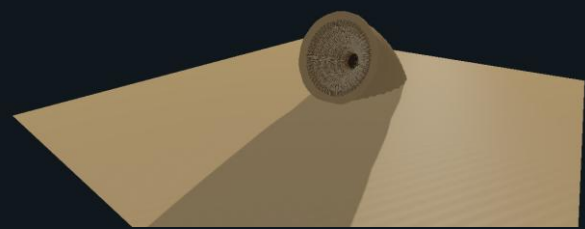
R/S: radial/tangential; r/d: row radius/depth.
l: reach; c: curl; D: drop; w: base width; t: base -> tip.

A readable equation-to-object mapping: dunes, hull/tank/worker curves, beacon rings, flexible worm and curved teeth.

Complex motion / One changing rescue scene

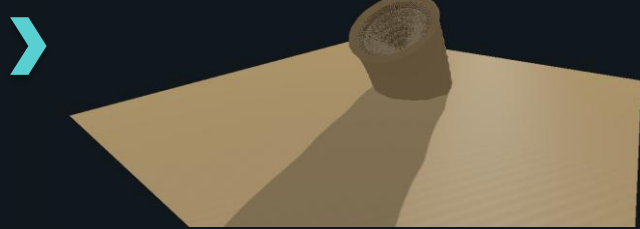
Objects move independently, while mission events change what the player can rescue and what remains visible.

APPROACH



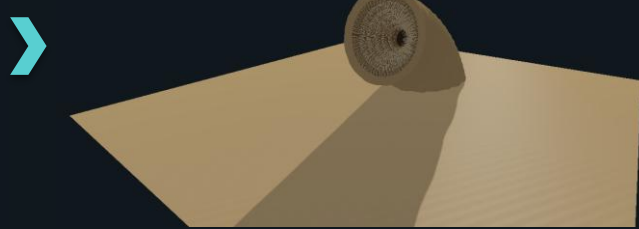
Wake + closing gap

RISE



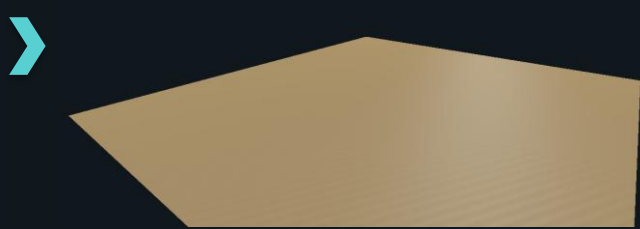
0-6 s after breach

ATTACK / SWALLOW



6-12 s: lunge + sink

RETREAT



13.5-18 s: withdraw

FLIGHT + ARTICULATION

$$V_{\text{next}} = v + (v_d - v)(1 - e^{-k\Delta t})$$

$$\beta_1 = 34\sin \phi, \quad \beta_2 = 34\sin(\phi + .75\pi)$$

k = 5.5 normal / 12 braking; speeds 28 / 48 units/s.
Terrain following + pitch/bank; analytically smoothed yaw.
Wing angles in degrees; phase 23 -> 36 rad/s; twist/exhaust.
Tracks/drum follow distance travelled, not idle clock time.

AUTONOMOUS ROUTE + EVACUATION

$$x = -40 + .62t + 15\sin(.045t)$$

$$z = -55 + 22\sin(.032t) + 9\sin(.071t)$$

t: seconds; tangent heading and slope alignment.
Groups release at 6 + 7j s; crew run at 3.8 units/s.
Worm gap: 220 -> 18; limbs animate; radius/height hazards.
Current breach: 150 s; then 60 s final extraction.

Game-like controls / Fly, inspect, rescue

The aircraft and the mission respond to input; changing view does not change the aircraft's heading.

FLIGHT INPUT

W / S	Forward / reverse thrust
A / D	Strafe without turning the nose
Q / E	Raise / lower terrain-following altitude
MOUSE	Hover steers yaw + camera pitch; centre dead zone
ARROWS	Left/right turn; up/down camera elevation
SHIFT	Rechargeable boost; release after depletion
SPACE	Brake + descend + winch / unload

Safety: altitude 4-55; rock/harvester clearance help.
Focus loss pauses and clears held input; reset/resize recentres the cursor. Movement stays inside +/-160.

C / THREE HEADING-FOLLOWING VIEWS



WIDE / 40 CLOSE / 20 TACTICAL / 100 UP

Smooth follow + terrain-safe eye; 58-degree perspective.
All are third-person, including the close preset.

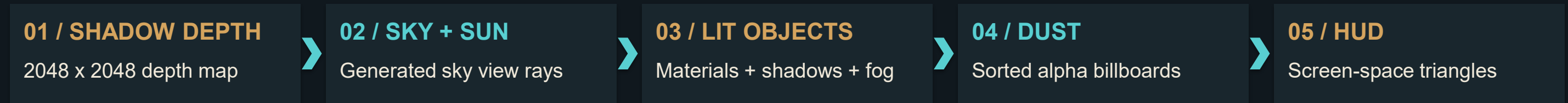
SPACE / RESCUE IS A STATE CHANGE

Pickup: distance <12, height <=11, speed <6.
Hold 0.42 s/person; cabin capacity = 8.
Deliver: base distance <14; hold 1.2 s per load.

Inside -> Running/Waiting -> Aboard -> Safe
Only Safe increases score; danger/timeout can cause Lost.

Rendering / What makes the scene convincing

A forward OpenGL renderer: one shadow target, then the visible frame, particles, and the interface.



SHADOWS + MATERIALS



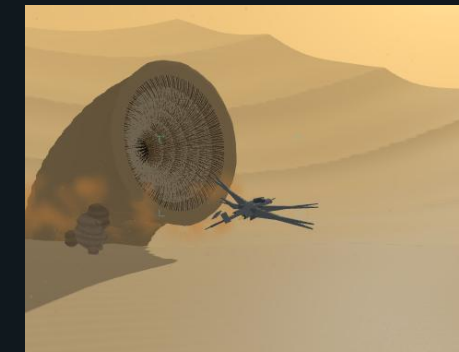
Directional Blinn-Phong + sand-wrap diffuse.
5 x 5 PCF; receiver-plane correction + bias.
Glossy opaque surfaces; emissive ornaments.

PROCEDURAL SURFACE DETAIL



Sand ripples, bump-like normals and sparkle.
Rock strata/grain; no bitmap texture files.
Sky gradient, sun disc/halo, distance fog, tone mapping.

DUST + COLLAPSING TERRAIN



BREACH DUST



GPU DEPRESSION

2,400 particles: wind/crawler/worm/downwash.
Sorted instances + lifetime/alpha fades.
Depth-tested, camera-facing billboards.
Crater: 13 units deep, tapers to radius 48.

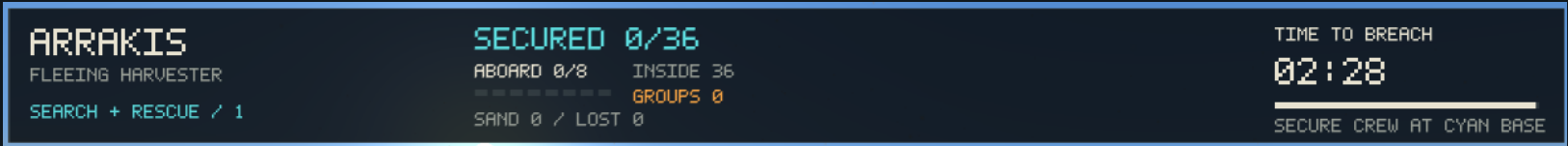
Depth/culling + requested 4x MSAA; inverse-transpose normals. Crater is GPU-only: CPU ground/collision height is unchanged.

DEMO: inspect long shadows + surface detail; press F for triangle topology; show breach dust and the visual depression.

Feedback / The player can read and hear the mission

Custom HUD, spatial sound, runtime safeguards, and reproducible build/testing tools.

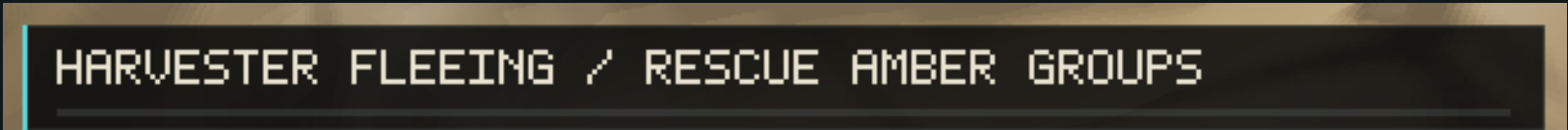
INSTRUMENTATION / 5 x 7 BITMAP FONT



Crew / cabin / score / wave; breach or extraction timer; seat and progress bars.



Altitude, speed, FPS, boost, threat gap.
North-up radar: cyan aircraft/base; amber crew;
white harvester; red worm; world range +/-300.
Adaptive layout and camera guide brackets.



Context prompts; title/pause/debrief panels; delivery-only score; session best.

8 SPATIAL SOUND EFFECTS

Loops: flight / boost / harvester / worm rumble.
Events: breach / winch / safe delivery / crew help.
Camera listener; distance attenuation; speed pitch;
boost/threat volume; mute and pause silencing.

Silent fallback; focus pause and cursor recentering.
Fullscreen/resize; minimized waiting; substeps.

ENGINEERING + CURRENT VERIFICATION

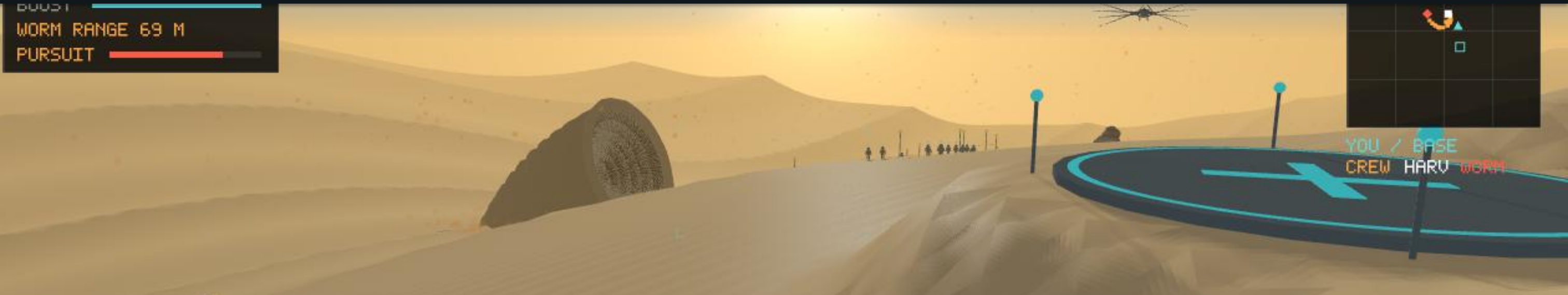
C++17 / GLFW / GLAD / GLM / miniaudio.
CMake + Visual Studio; audio copied post-build.
Procedural startup meshes and embedded GLSL.
Report PDF + source documentation/manifest tooling.

4 / 4 graphical smoke modes passed

23 / 27 mission cases; 1,057 checks

Current timer is 150 s; five assertions fail because
tests expect 210-to-150 progression. Source untouched.

THANK YOU



Procedural geometry. Mathematical motion. A controllable rescue mission.

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Objects, curved equations, dynamic scenes, cameras, interaction, shadows, surface detail, dust, HUD and sound.

Project report available. Your completed demonstration video is separate; no video is embedded in this deck.
Limits: arcade physics; opaque glass; GPU-only crater. 'Karpov objects' is undefined and is not claimed.